Assignment No 6

Title

Capture and Analyze the TLS Handshake Using Wireshark and Verify the Digital Certificate Used for Secure Communication

1. Aim

To capture and analyze a **TLS handshake** using Wireshark and verify the digital certificate used to establish secure communication between a client and server.

2. Objectives

1. To understand the purpose of TLS in secure communication.
2. To capture TLS packets using Wireshark.
3. To study the different stages of the TLS handshake.
4. To examine the digital certificate presented by the server.
5. To verify certificate validity, issuer, subject, and expiration information.
6. To understand how TLS protects data from interception and modification.

3. Introduction

Transport Layer Security (TLS) is a cryptographic protocol used to secure communication between applications over a network. It provides **confidentiality, integrity, and authentication**. HTTPS is a common example of an application protocol that uses TLS to protect web communication.

Before encrypted application data is exchanged, the client and server perform a **TLS handshake**. During this process, they negotiate security parameters, authenticate the server using a digital certificate, and establish cryptographic keys for subsequent communication.

Wireshark can capture and display TLS handshake packets, allowing security professionals to study the connection establishment process without exposing the encrypted application data itself.

4. TLS Handshake

A simplified TLS handshake involves the following stages:

4.1 ClientHello

The client initiates the connection by sending a **ClientHello** message. It may contain:

- Supported TLS versions
- Random value
- Supported cipher suites
- Extensions

- Server Name Indication (SNI)

4.2 ServerHello

The server responds with a **ServerHello** message and selects the parameters that will be used for the connection, such as the TLS version and cryptographic algorithms.

4.3 Server Certificate

The server sends its **digital certificate** to the client. The certificate contains information such as:

- Subject
- Issuer
- Validity period
- Public key
- Signature
- Certificate extensions

The certificate allows the client to authenticate the identity of the server.

4.4 Key Establishment

The client and server perform a cryptographic key-exchange procedure. Modern TLS commonly uses ephemeral key exchange mechanisms to establish shared session secrets.

4.5 Finished Messages

After the required cryptographic parameters have been established, the client and server exchange **Finished** messages to confirm that the handshake was completed successfully.

After this stage, application data can be transmitted through the encrypted TLS connection.

5. Wireshark Analysis

Wireshark can be used to identify TLS packets within a captured network trace. A display filter such as:

tls

can be used to focus on TLS traffic.

Important packets to identify include:

- Client Hello
- Server Hello
- Certificate
- Key exchange-related messages
- Finished

The packet details can be expanded in Wireshark to examine the TLS version, cipher suite, extensions, and certificate information.

6. Digital Certificate Verification

The server certificate should be examined to determine whether it is valid and trustworthy.

Important certificate parameters include:

Subject:
Identifies the entity for which the certificate was issued.

Issuer:
Identifies the Certificate Authority (CA) that issued the certificate.

Validity Period:
Specifies the certificate's start and expiration dates.

Public Key:
Contains the public key associated with the certificate.

Signature Algorithm:
Indicates the algorithm used by the issuer to sign the certificate.

Subject Alternative Name (SAN):
Specifies the domain names or identities for which the certificate is valid.

Certificate Chain:
Shows the relationship between the server certificate and the trusted CA hierarchy.

7. Certificate Security Checks

A certificate should satisfy the following conditions:

1. It should not be expired or not-yet-valid.
2. The requested domain should match an appropriate **SAN** entry.
3. It should be issued by a trusted Certificate Authority.
4. The certificate chain should validate correctly.
5. Weak or obsolete cryptographic algorithms should be avoided.
6. The certificate should not be revoked when revocation checking is applicable.

A valid certificate alone does not guarantee that the entire connection is secure; the TLS version, negotiated cryptographic algorithms, endpoint configuration, and application behavior should also be considered.

8. Security Importance of TLS

TLS provides three major security properties:

Confidentiality:
Encryption prevents unauthorized parties from reading application data.

Integrity:
Cryptographic protection helps detect unauthorized modification of transmitted data.

Authentication:
Digital certificates allow the client to verify the identity of the server.

Without properly configured TLS, attackers may potentially intercept, modify, or impersonate network services.

9. Observation

During TLS traffic analysis, the handshake packets can be identified in Wireshark. The ClientHello and ServerHello messages provide information about the negotiated TLS parameters, while the Certificate message provides information about the server's digital certificate.

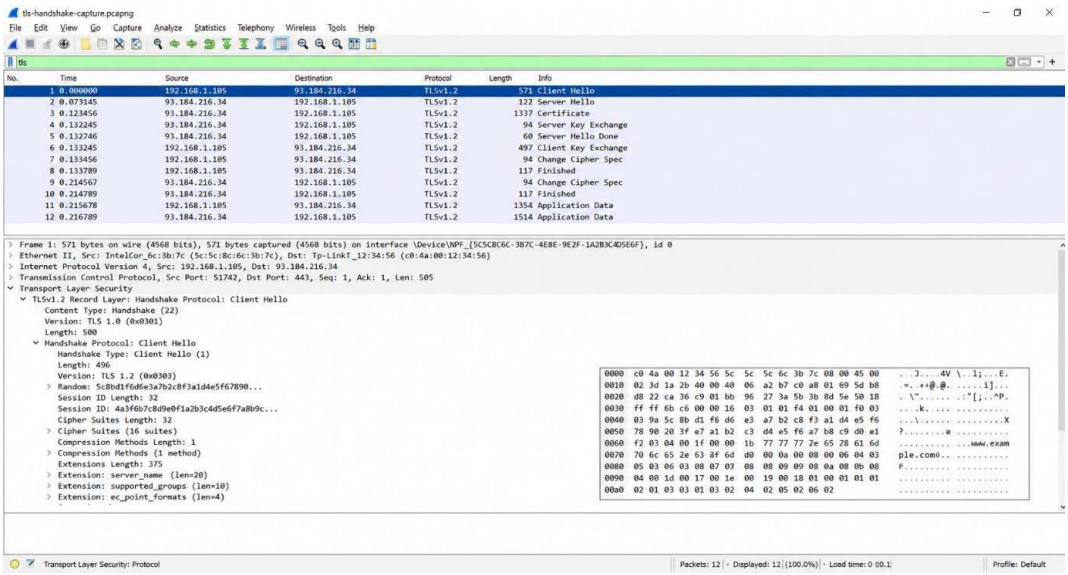
The application data transmitted after the successful handshake is encrypted, demonstrating the role of TLS in protecting network communication.

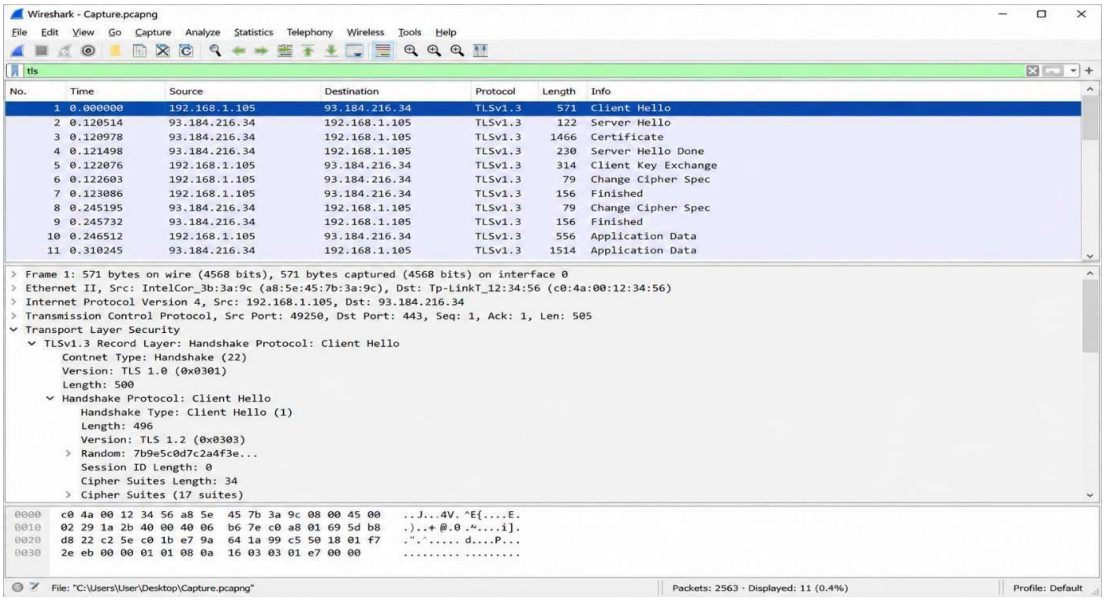
10. Result

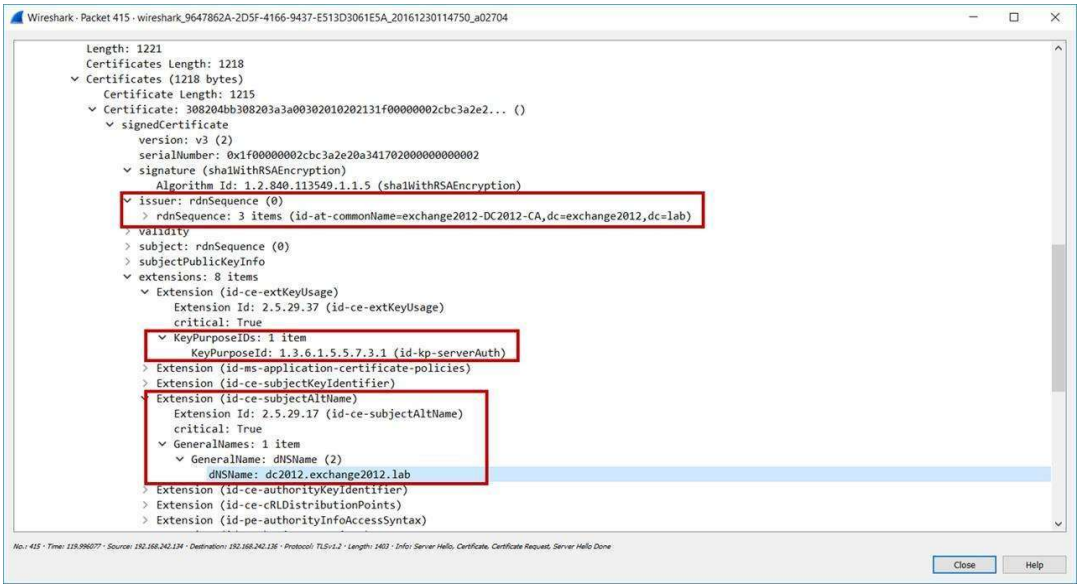
The TLS handshake was analyzed using Wireshark, and the server's digital certificate was examined. The certificate's subject, issuer, validity period, public-key information, and certificate chain can be used to verify the identity and trustworthiness of the secure connection.

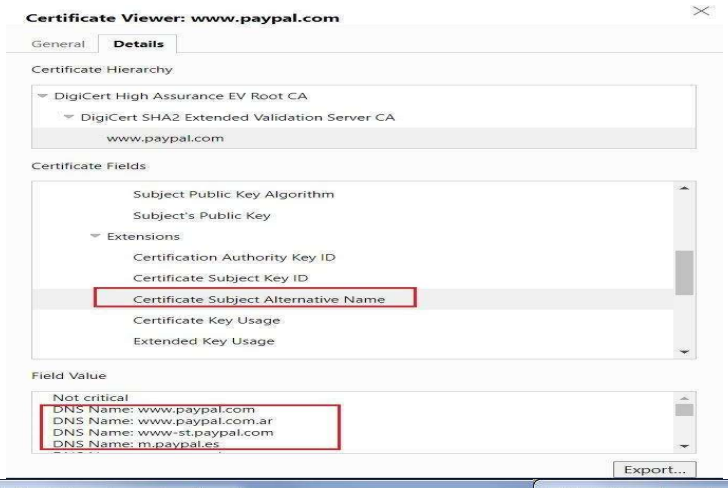
11. Conclusion

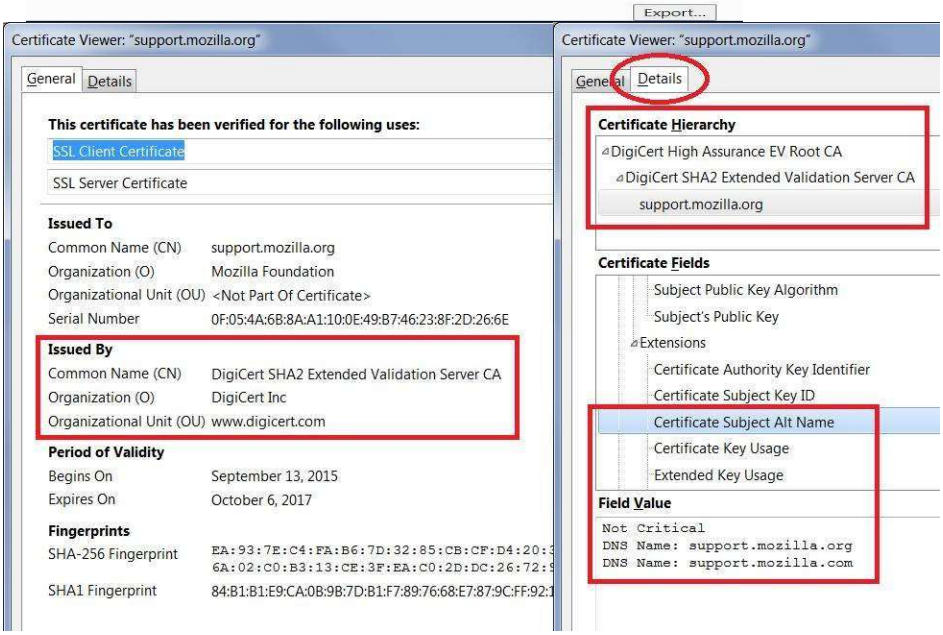
The TLS handshake is an essential process for establishing secure communication between a client and server. Wireshark provides useful visibility into the handshake messages and negotiated security parameters. Verification of the server's digital certificate helps establish server identity and trust.




Wireshark - Capture.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tls
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.105 93.184.216.34 TLSv1.3 571 Client Hello
2 0.120514 93.184.216.34 192.168.1.105 TLSv1.3 122 Server Hello
3 0.120978 93.184.216.34 192.168.1.105 TLSv1.3 1466 Certificate
4 0.121498 93.184.216.34 192.168.1.105 TLSv1.3 230 Server Hello Done
5 0.122076 192.168.1.105 93.184.216.34 TLSv1.3 314 Client Key Exchange
6 0.122603 192.168.1.105 93.184.216.34 TLSv1.3 79 Change Cipher Spec
7 0.123086 192.168.1.105 93.184.216.34 TLSv1.3 156 Finished
8 0.245195 93.184.216.34 192.168.1.105 TLSv1.3 79 Change Cipher Spec
9 0.245732 93.184.216.34 192.168.1.105 TLSv1.3 156 Finished
10 0.246512 192.168.1.105 93.184.216.34 TLSv1.3 556 Application Data
11 0.310245 93.184.216.34 192.168.1.105 TLSv1.3 1514 Application Data
Frame 1: 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface 0
Ethernet II, Src: IntelCor_3b:3a:9c (a8:5e:45:7b:3a:9c), Dst: Tp-LinkT_12:34:56 (c0:4a:00:12:34:56)
Internet Protocol Version 4, Src: 192.168.1.105, Dst: 93.184.216.34
Transmission Control Protocol, Src Port: 49250, Dst Port: 443, Seq: 1, Ack: 1, Len: 505
Transport Layer Security
TLSv1.3 Record Layer: Handshake Protocol: Client Hello
Contnet Type: Handshake (22)
Version: TLS 1.0 (0x0301)
Length: 500
Handshake Protocol: Client Hello
Handshake Type: Client Hello (1)
Length: 496
Version: TLS 1.2 (0x0303)
Random: 7b9e5c0d7c2a4f3e...
Session ID Length: 0
Cipher Suites Length: 34
Cipher Suites (17 suites)
0000 c0 4a 00 12 34 56 a8 5e 45 7b 3a 9c 08 00 45 00 ..J..4V. ^E{....E.
0010 02 29 1a 2b 40 00 40 06 b6 7e c0 a8 01 69 5d b8 .)..+@.@. .~....i].
0020 d8 22 c2 5e c0 1b e7 9a 64 1a 99 c5 50 18 01 f7 ."..^.... d...P...
0030 2e eb 00 00 01 01 08 0a 16 03 03 01 e7 00 00 ....
File: "C:\Users\User\Desktop\Capture.pcapng"
Packets: 2563 · Displayed: 11 (0.4%)
Profile: Default


Wireshark · Packet 415 · wireshark_9647862A-2D5F-4166-9437-E513D3061E5A_20161230114750_a02704
Length: 1221
Certificates Length: 1218
Certificates (1218 bytes)
Certificate Length: 1215
Certificate: 308204bb308203a3a003020102021 31f00000002cbc3a2e2... ()
signedCertificate
version: v3 (2)
serialNumber: 0x1f00000002cbc3a2e20a34170200000000002
signature (sha1WithRSAEncryption)
Algorithm Id: 1.2.840.113549.1.1.5 (sha1WithRSAEncryption)
issuer: rdnSequence (0)
rdnSequence: 3 items (id-at-commonName=exchange2012-DC2012-CA,dc=exchange2012,dc=lab)
validity
subject: rdnSequence (0)
subjectPublicKeyInfo
extensions: 8 items
Extension (id-ce-extKeyUsage)
Extension Id: 2.5.29.37 (id-ce-extKeyUsage)
critical: True
KeyPurposeIDs: 1 item
KeyPurposeId: 1.3.6.1.5.5.7.3.1 (id-kp-serverAuth)
Extension (id-ms-application-certificate-policies)
Extension (id-ce-subjectKeyIdentifier)
Extension (id-ce-subjectAltName)
Extension Id: 2.5.29.17 (id-ce-subjectAltName)
critical: True
GeneralNames: 1 item
GeneralName: dNSName (2)
dNSName: dc2012.exchange2012.lab
Extension (id-ce-authorityKeyIdentifier)
Extension (id-ce-cRLDistributionPoints)
Extension (id-pe-authorityInfoAccessSyntax)
No.: 415 · Time: 119.996077 · Source: 192.168.242.134 · Destination: 192.168.242.136 · Protocol: TLSv1.2 · Length: 1403 · Info: Server Hello, Certificate, Certificate Request, Server Hello Done
Close
Help


Certificate Viewer: www.paypal.com
General
Details
Certificate Hierarchy
DigiCert High Assurance EV Root CA
DigiCert SHA2 Extended Validation Server CA
www.paypal.com
Certificate Fields
Subject Public Key Algorithm
Subject's Public Key
Extensions
Certification Authority Key ID
Certificate Subject Key ID
Certificate Subject Alternative Name
Certificate Key Usage
Extended Key Usage
Field Value
Not critical
DNS Name: www.paypal.com
DNS Name: www.paypal.com.ar
DNS Name: www-st.paypal.com
DNS Name: m.paypal.es
Export...


Certificate Viewer: "support.mozilla.org"
General
Details
This certificate has been verified for the following uses:
SSL Client Certificate
SSL Server Certificate
Issued To
Common Name (CN) support.mozilla.org
Organization (O) Mozilla Foundation
Organizational Unit (OU) <Not Part Of Certificate>
Serial Number 0F:05:4A:6B:8A:A1:10:0E:49:B7:46:23:8F:2D:26:6E
Issued By
Common Name (CN) DigiCert SHA2 Extended Validation Server CA
Organization (O) DigiCert Inc
Organizational Unit (OU) www.digicert.com
Period of Validity
Begins On September 13, 2015
Expires On October 6, 2017
Fingerprints
SHA-256 Fingerprint
SHA1 Fingerprint
Certificate Viewer: "support.mozilla.org"
General
Details
Certificate Hierarchy
DigiCert High Assurance EV Root CA
DigiCert SHA2 Extended Validation Server CA
support.mozilla.org
Certificate Fields
Subject Public Key Algorithm
Subject's Public Key
Extensions
Certificate Authority Key Identifier
Certificate Subject Key ID
Certificate Subject Alt Name
Certificate Key Usage
Extended Key Usage
Field Value
Not Critical
DNS Name: support.mozilla.org
DNS Name: support.mozilla.com